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Association between intra and inter-limb strength asymmetry with sprint kinematics and force-velocity profile in youth team athletes. The Knee, 52: 99-107, 2025.

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Does pad displacement correlate to impact severity metrics in football helmets? Journal of Sports Engineering and Technology, Ahead of Print: 1-13, 2024.

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Forces experienced at different levels of the musculoskeletal system during horizontal decelerations. Journal of Sports Sciences, Ahead of Print: 1-13, 2024.

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Increased anterior pelvic tilt angle elevates the risk of hamstring injuries in soccer player. Research in Sports Medicine, Ahead of Print: 1-18, 2024.

Influence of countermovement depth on net force, push-off time, vertical impulse and performance during jumping. Journal of Electromyography and Kinesiology, 79: 102945, 2024.

Influence of interlimb lean muscle mass asymmetry on countermovement jump neuromuscular performance qualities among American football players. Journal of Strength and Conditioning Research, Ahead of Print: 1-7, 2024.

Multiscale hamstring muscle adaptations following 9 weeks of eccentric training. Journal of Sport and Health Science, Ahead of Print: 1-10, 2024.

Platelet-poor plasma for the treatment of acute hamstring muscle injuries in collegiate football athletes: A cohort study. Clinical Journal of Sport Medicine, Ahead of Print: 1-5, 2024.

Refining the recipe or spoiling the broth? Reframing perceptions of program adaptation in sports injury prevention. British Journal of Sports Medicine, Ahead of Print: 1-2, 2024.

Repeated sprint training: The effects of session volume on acute physiological, neuromuscular, perceptual and performance outcomes in athletes. European Journal of Sport Science, Ahead of Print: 1-13, 2024.

Return to play after shoulder stabilization – What if this isn't going well? Operative Techniques in Sports Medicine, Ahead of Print: 1-24, 2024.

Return to play in professional football players following traumatic cervical spine injury: Expert opinions from the National Football League spine surgeons. JNS Spine, Online: Dec. 20: 1-10, 2024.

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Risk and prognosis of hamstring injuries in the National Football League: A 12-year review. The Orthopaedic Journal of Sports Medicine, 12(12): 1-5, 2024.

The effect of multi-ingredient protein versus collagen supplementation on satellite cell properties in males and females. Medicine and Science in Sports and Exercise,56(11): 2125-2134, 2024.

The effect of the Copenhagen Adduction and Nordic Hamstring exercises on muscle strength and athletic performance: A quasi-experimental study. Journal of Orthopaedic Reports, Ahead of Print: 1-42, 2024.

The effect of the Nordic Hamstring exercise on hamstring muscle activity distribution during high-speed running estimated using multichannel electromyography: A pragmatic randomized controlled trial. Clinical Journal of Sport Medicine, Ahead of Print: 1-10, 2024.

The effectiveness of cryostimulation exposure on sleep and recovery in male athletes: Timing considerations. European Journal of Sport Science, Ahead of Print: 1-10, 2024.

The potential role of retinal slip on National Football League (NFL) officiating and its effect on on-field decision making: Mechanisms and countermeasures. The Royal College of Ophthalmologists, Ahead of Print: 1-6, 2024.

The role of speed, change of direction, and momentum by position and starting status in Division 1 collegiate football players. Journal of Strength and Conditioning Research, 39(1): 41-47, 2024.

The state of the science for potential contributors to musculoskeletal injury following concussion: Mechanisms, gaps, and clinical considerations. Musculoskeletal Science and Practice, 75: 103219, 2025.

The structure, function, and adaptation of lower-limb aponeuroses: Implications for myo-aponeurotic injury. Sports Medicine – Open, 10: 133, 2024.

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Theorizing the use of performance enhancing substances and methods in sport as four different types of deviant behavior. Deviant Behavior, Ahead of Print: 1-19, 2024.

3.0-Tesla MRI observation at return to play after hamstring injuries. Clinical Journal of Sport Medicine, Ahead of Print: 1-8, 2024.

Objective: To describe 3.0-Tesla (T) magnetic resonance imaging (MRI) findings of hamstring muscles in clinically recovered athletes who were cleared for return to play (RTP).

Design: Prospective observational study.

Setting: Hospital.

Participants: Athletes (amateur and professional) who were cleared for RTP after hamstring injury. Fifty-eight participants were included in the analysis.

Independent variables: 3-T MRI at baseline (within 7 days from initial injury) and MRI at RTP (within 10 days of RTP).

Main outcome measures: Injury location, grade of injury (modified Peetrons and British Athletics Muscle Injury Classification/BAMIC), presence and the extent of intramuscular signal abnormality, intramuscular tendon disruption, and thickness. Reinjuries within 1 year of RTP were recorded.

Results: Magnetic resonance images at RTP showed that 55 (95%) participants had intramuscular increased signal intensity (edema) and 44 (76%) participants had intramuscular abnormal low-signal intensity (suggesting fibrosis) on MRI. There was an overall reduction of injury grades according to the modified Peetrons and BAMIC classification at initial injury to RTP. Three (5%) participants had no abnormal signal intensities (grade 0 or grade 0A) on MRI at RTP. Intramuscular tendon disruption, waviness, and tendon thickening were present at RTP in, respectively, 22 (38%), 15 (26%), and 36 (62%). We recorded 3 (5%) reinjuries.

Conclusions: At RTP, 3.0-T MRI shows high percentages of MRI abnormalities (edema, fibrosis, and intramuscular tendon disruption and thickening). We conclude that complete normalization of 3.0-T MRI

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is not expected for RTP decision after a hamstring injury. Its possible association with reinjury risk has to be determined in larger cohorts.

A new field-test for assessing the medial and lateral hamstring strength at long-muscle length. Journal of Bodywork & Movement Therapies., 41: 78-84, 2025.

Abstract - This research investigates the influence of tibial rotations with knee flexion (KF) on the electromyographic (EMG) activities of hamstring muscles (HM) groups and the strength ratio between the medial and lateral rotation of the tibia. A cross-sectional design was employed to assess muscle activity, isometric strength and reliability. The research was conducted in a controlled laboratory environment. Thirty-six amateur male athletes were recruited as volunteers. The measures included peak muscle activity of the semitendinosus and biceps femoris long head, the knee flexors' isometric strength ratio and reliability. The isometric strength ratios of medial (MR) to lateral (LR) tibial rotations were 0.94 ± 0.17 at 90°, and 0.93 ± 0.10 at both 60° and 30° of KF angulation. Tibial position significantly influenced knee flexion strength as well as HM activity, irrespective of KF angulation. Specifically, biceps femoris activity increased by 33.6% in LR compared to MR, while semitendinosus activity increased by 22.6% in MR compared to LR. The Knee-Rotation test (KR-test) can be a valuable tool for evaluating both HM groups (ICC >0.87), and identifying the primary target for strengthening purposes during the injury prevention process. It provides insights for effective rehabilitation and training interventions.

A practitioner's guide to isometric hip adduction and abduction test selection: Maximal force, asymmetry, and muscle activity comparisons between assessment positions. Journal of Sports Sciences, Ahead of Print: 1-9, 2024.

Abstract - The purpose of this study was to compare maximal adduction (ADD) and abduction (ABD) force, ADD to ABD ratio (ADD:ABD), inter-limb asymmetries and muscle activity between five isometric hip strength assessment positions. Twenty male athletes performed the following positions: seated (SE), supine at knees with 0° hip flexion (SK0°) and 45° hip flexion (SK45°), and supine at ankles bilaterally (SABI) and unilaterally (SAUNI). Normalised muscle activity (%EMGmax) of adductor longus (ADDLONG),

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gracilis (GRAC), gluteus medius (GMED), sartorius (SAR), and lower rectus abdominis (RECABD) were also assessed. Significant main effects of assessment position were identified for relative peak force (rPF) and torque (rT), ADD:ABD, ADD asymmetry, and %EMGmax. Significantly greater ADD and ABD rPF and rT were observed in SE and SK45° compared to SABI and SAUNI. SK45° elicited the highest ADDLONG and GRAC %EMGmax during ADD, with SE and SK45° evoking the highest GMED %EMGmax during ABD. SK0° exhibited significantly higher ADD:ABD compared to all positions except SAUNI. ADD asymmetry was significantly largest in SAUNI, and RECABD %EMGmax was highest in SABI during ADD. This data can inform selection of assessment positions based on whether the primary measures of interest are maximal force and muscle activity, or identifying asymmetries.

Association between intra and inter-limb strength asymmetry with sprint kinematics and force-velocity profile in youth team athletes. The Knee, 52: 99-107, 2025.

Background: Maximal force plays a pivotal role in enhancing performance across various dynamic sports, particularly in sprinting biomechanics. However, muscle strength asymmetry among major muscle groups like the hamstrings and quadriceps may hinder sprint performance and raise injury risks. This study examines how intra- and inter-limb strength asymmetry relates to sprint kinematics and the power-force-velocity profile in youth athletes.

Methods: Seventy-four youth athletes from football, volleyball, and handball completed two testing sessions, 48 h apart. Assessments included anthropometric measurements, sprint tests, and isokinetic dynamometer strength evaluations at 60°.s-1 and 180°.s-1. Sprint kinematics were analyzed using the Optojump Next system, and force-velocity profile variables were determined via Samozino's method. Asymmetry analysis used a standard percentage difference equation.

Results: Weak to moderate correlations (r = -0.46 to 0.45) were found between strength asymmetry and sprint parameters, suggesting limited impact of asymmetry on sprint kinematics and kinetics. Although

most participants showed asymmetry levels below the 15% threshold deemed functionally significant, outliers had asymmetry values up to 42%, indicating considerable variability.

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Conclusions: Lower extremity strength asymmetry moderately affects sprint kinematics and force-velocity profiles in young team athletes, challenging traditional views. Further research is needed to explore the mechanisms between strength asymmetry and sprint performance, aiming to inform targeted training interventions that optimize performance and reduce injury risks in youth athletes.

Biceps femoris long head fascicle length not associated with hamstring injury risk in NCAA football athletes. Clinical Journal of Sport Medicine, Ahead of Print: 1-3, 2024.

Objective: To determine whether biceps femoris long head (BFlh) fascicle length (FL) is related to hamstring injury (HSI) risk in collegiate football athletes.

Design: Retrospective cohort.

Setting: University athletic medical center.

Participants: Division I Collegiate football athletes (n = 129, age = 20.03 ± 1.68 years, body mass index = 29.50 ± 5.06 kg/m^2).

Interventions: Biceps femoris long head FL measurements were previously taken on the nondominant leg of each athlete. Presence or absence of HSI for 2 playing seasons and training camps was evaluated through retrospective review of medical records. Main

Outcome Measures: Whether a shorter BFlh FL was associated with increased risk for any HSI.

Results: Biceps femoris long head FL was not associated with HSI risk. Body mass index $30 kg/m2 was incidentally found to be protective (odds ratio 5 0.27, 95% confidence interval, 0.08–0.84, P 5 0.03).

Conclusions: Biceps femoris long head FL is not an ideal measurement for predicting risk of HSI in American football athletes.

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Does pad displacement correlate to impact severity metrics in football helmets? Journal of Sports Engineering and Technology, Ahead of Print: 1-13, 2024.

Abstract - As efforts to reduce sport-related head injuries continue, there is a need to better understand the mechanisms of an impact that influence the risk of sustaining a concussion. This study presents a method which can be used to separate football helmet pad displacement into normal and rotational components. Finite element models of three popular football helmets were used to simulate a variety of impacts per the National Football League (NFL) helmet testing protocol. Pad displacement components were evaluated for correlation with impact severity metrics including Head Injury Criterion (HIC), Diffuse Axonal Multi-Axis General Evaluation (DAMAGE), and Head Acceleration Response Metric (HARM). When combining all locations, displacement was not found to correlate well with severity metrics ($R^2 < 0.44$). However, trends did exist with normal displacement when evaluating data by impact location. Normal displacement was found to correlate strongly with HARM, HIC, and Peak Resultant Linear Acceleration (PRLA) at the side upper (SU) location ($R^2 = 0.79–0.8$). Overall, it was observed that helmets which maximize normal displacement without bottoming out performed better. Tangential displacement was not found to correlate well with any impact severity metrics even when evaluated by location. This study has provided a method to evaluate components of helmet pad normal displacement during simulations which may help guide design and optimization efforts for helmets under impact in critical locations.

Eccentric hamstring strength imbalance among football and soccer athletes. The Southern Medical Association, 117(4): 214-219, 2024.

Objectives: Hamstring strain injuries (HSI) are common among football and soccer athletes. Eccentric strength imbalance is considered a contributing factor for HSI. There is, however, a paucity of data on hamstring imbalances of soccer and American football athletes as they age and advance in skill level. High school athletes will display greater interlimb discrepancies compared with collegiate and professional athletes. In addition, soccer athletes will exhibit greater hamstring asymmetry than American football athletes.

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Methods: Hamstring testing was performed on soccer and American football athletes using the NordBord Hamstring Testing System (Vald Performance, Albion, Australia). Age, sex, weight, sport specialization, and sport level were recorded. Maximum hamstring forces (N), torque (N · m), and work (N · s) were measured. Hamstring imbalance (%) was calculated by dividing the absolute value of the difference in leg forces divided by their sum. One-way analysis of variance and independent sample t tests compared measurements between athlete groups.

Results: A total of 631 athletes completed measurements, including 88 high school male soccer, 25 college male soccer, 23 professional male soccer, 83 high school female soccer, 28 college female soccer, 288 high school football, and 96 college football athletes. High school soccer players displayed significantly greater imbalances for torque ($P = 0.03$) and work ($P < 0.01$) than football athletes. Imbalances for maximum force ($P = 0.035$), torque ($P = 0.018$), and work ($P = 0.033$) were significantly higher for male soccer athletes in high school compared with college- and professional-level athletes. Female high school soccer players had significantly higher imbalance in torque ($P = 0.045$) and work ($P = 0.001$) compared with female collegiate soccer players. Football athletes did not experience significant changes in force imbalances between skill levels.

Conclusions: High school soccer athletes exhibit greater hamstring imbalances than football athletes. Higher levels of play in soccer, for both male and female athletes, correlate with less hamstring asymmetry.

Effects of prematch travel and training on physical and technical performance in international football matches. International Journal of Sports Physiology and Performance, Ahead of Print: 1-7, 2024.

Purpose: This study examined the relationship of prematch travel and in-camp training on ensuing physical and technical match performance of footballers (soccer) competing for a national team.

Methods: Match running and technical performance data were obtained from 68 national-team footballers competing in international matches (N = 108). Match performance data were aligned with the confirmed travel durations, time-zone change, travel direction, and time between arrival and kickoff

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for the travel to the match. In addition, in-camp training load from the 3 days prior to national-team matches was also collated. Linear mixed models assessed relationships between travel and training measures with physical and technical match performance outcomes.

Results: Travel variables explained little variance in outcomes (R2 = .02-.16). Traveling eastward was associated with an increase in total (P = .042) and very high-speed distance (P = .030) in matches and a 5% decrease in pass accuracy (P = .012). Greater time-zone difference was associated with increased match decelerations (P = .027), while arriving later for matches was associated with increased total (P = .041) tackles. Increases in training high-speed distance were associated with increases in match high-speed distance (P = .004) and number of decelerations (P < .001). An increase in training total distance was associated with a decrease in match decelerations (P = .002).

Conclusions: Prematch travel appeared to have minimal effects on physical and technical performance in this national team. Training loads prior to matches may have some relationship with match running performance, and, thus, readiness to train should be a priority for athletes arriving in national-team camps.

Forces experienced at different levels of the musculoskeletal system during horizontal decelerations. Journal of Sports Sciences, Ahead of Print: 1-13, 2024.

Abstract - Horizontal decelerations are frequently performed during team sports and are closely linked to sports performance and injury. This study aims to provide a comprehensive description of the kinetic demands of decelerations at the whole-body, structural, and tissue-specific levels of the musculoskeletal system. Team-sports athletes performed maximal-effort horizontal decelerations whilst full-body kinematics and ground reaction forces (GRFs) were recorded. A musculoskeletal model was used to determine whole-body (GRFs), structural (ankle, knee, and hip joint moments and contact forces), and tissue (twelve lower-limb muscle forces) loads. External GRFs in this study, especially in the horizontal direction, were up to six times those experienced during accelerated or constant-speed running reported in the literature. To cope with these high external forces, large joint moments (hip immediately after touchdown; ankle and knee during mid and late stance) and contact forces (ankle, knee, hip immediately after touchdown) were observed. Furthermore, eccentric force requirements of the tibialis

anterior, soleus, quadriceps, and gluteal muscles were particularly high. The presented loading patterns provide the first empirical explanations for why decelerating movements are amongst the most challenging in team sports and can help inform deceleration-specific training prescription to target horizontal deceleration performance, or fatigue and injury resistance in team-sports athletes.

Has COVID-19 led to more sudden cardiac deaths in football? Clinical Research in Cardiology, Online: Nov. 24: 1-5, 2024.

Introduction: It is unclear whether the number of sudden cardiac death (SCD) and survived sudden cardiac arrest (SCA) has increased among football players during the COVID-19 pandemic. This study aims to compare the SCD/SCA burden between the pre-pandemic period and COVID-19 pandemic in football players worldwide.

Methods: The COVID-19 pandemic and an equivalent pre-pandemic period (each lasting 1151 days) were analyzed for SCD/SCA by extracting data from the prospective FIFA (Fédération Internationale de Football Association) Sudden Death Registry. Particular focus was placed on cardiac diseases acquired through the novel coronavirus SARS-CoV-2, such as myocarditis and coronary artery disease (CAD), potentially leading to SCD/SCA.

Results: There were 454 SCD/SCA (survival rate: 24%) and 380 SCD/SCA (survival rate: 27%) during the pre-pandemic period and COVID-19 pandemic, respectively ($p = 0.27$). In the pre-pandemic period, out of 191 confirmed and suspected diagnoses, there were 6 (3%) cases of myocarditis and 69 (36%) cases of CAD and during the pandemic out of 136 confirmed and suspected diagnoses, there was 1 (1%) case of myocarditis and 58 (43%) cases of CAD.

Conclusion: The burden of SCD/SCA, particularly myocarditis and CAD, in football players worldwide seemingly has not been higher during the COVID-19 pandemic than during a comparable period before.

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Increased anterior pelvic tilt angle elevates the risk of hamstring injuries in soccer player. Research in Sports Medicine, Ahead of Print: 1-18, 2024.

Objective: This study aims to determine the impact of increased anterior pelvic tilt angle on hamstring muscle injuries in football players. An independent t-test was performed to compare right and left anterior pelvic tilt angles.

Design: The study was conducted by monitoring 76 different players from a professional club in the Premier Division of Professional Football League over a period of five years. During this time, the hamstring injuries sustained by the players and their severity were recorded using MRI imaging techniques.

Result: The comparison of the right and left anterior pelvic tilt angles among players was analysed using an independent t-test for independent groups. The Receiver Operating Characteristic (ROC) analysis was used to determine the threshold value that categorizes the right and left angles. According to the results, a significant impact of the variation in right-left anterior pelvic tilt angle on hamstring muscle injuries was observed ($p < 0.05$). However, no significant effects were found for the Q angle, tibial torsion, or knee posture ($p > 0.05$).

Conclusion: In conclusion, it can be stated that an increase in the anterior pelvic tilt angle is associated with a higher frequency of hamstring injuries.

Influence of countermovement depth on net force, push-off time, vertical impulse and performance during jumping. Journal of Electromyography and Kinesiology, 79: 102945, 2024.

Abstract - The purpose of this study was to investigate the influence of push-off distance on maximal jump height and to characterize the relationships between vertical jump mechanical parameters. Seventeen CrossFit athletes performed maximal countermovement jumps on two force-plates, with four different push-off distances induced by different countermovement depths. Results confirmed that push-off distance and maximal jump height were related by a quadratic relationship, and that exists an

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individual push-off distance maximizing maximal jump height. Furthermore, the increase in push-off distance was associated with an increase in the time of push-off and a concomitant decrease in the mean vertical external force, changing the relative net vertical impulse and so maximal jump height. An almost perfect negative linear relationship was observed between mean vertical external force and time of push-off ($r = -0.99$, $p = 0.006$), mathematically resulting in a quadratic relationship between relative net vertical impulse and time of push-off ($r = 1.00$, $p < 0.001$). This negative linear relationship could be explained by changes in joint angles and velocities associated with an increase in push-off distance, modifying joint torque production capacities during vertical jumping. This provides new insights to characterize vertical jump mechanical parameters and ways to optimize training and jump performance.

Influence of interlimb lean muscle mass asymmetry on countermovement jump neuromuscular performance qualities among American football players. Journal of Strength and Conditioning Research, Ahead of Print: 1-7, 2024.

Abstract - Body composition is a determinant of athletic performance. The purpose of this study was to determine the influence of lower limb lean muscle mass (LMM) asymmetry on countermovement jump (CMJ) performance and kinetic asymmetry during both eccentric and concentric CMJ phases. Seventy-four American football players (age, 23.0 ± 0.9 years; height, 1.86 ± 0.07 m; mass, 104.9 ± 21.6 kg; lean body mass, 88.7 ± 12.5 kg) training for the 2024 National Football League draft underwent body composition assessment using an InBody 270 bioelectrical impedance analyzer that yielded right and left leg LMM in kg. Subjects then performed 3 CMJs on dual uniaxial force plates that allowed for the calculation of jump height (JH), reactive strength index modified (RSImod), and interlimb percentage asymmetry of peak force and impulse for eccentric and concentric phases. A paired t-test and Pearson Product-moment Correlation Coefficients calculated the difference between left and right LMM and the associations between the symmetry variables, respectively, $p \leq 0.05$. Right and left leg LMM were significantly different ($t(73) = 3.772$, $p \leq 0.001$). However, no significant relationships were found between lower limb LMM asymmetry (0.67 ± 0.67%) and any asymmetry in CMJ eccentric peak force (4.77 ± 3.75%), $r(72) = 0.001$, $p = 0.954$; eccentric deceleration impulse (5.23 ± 4.36%), $r(72) = 0.158$, $p = 0.180$; concentric peak force (2.99 ± 2.54%), $r(72) = -0.70$, $p = 0.553$; concentric impulse (3.06 ± 2.97%), $r(72) = 111$, $p = 0.347$; JH (41.8 ± 7.3 cm), $r(72) = -0.205$, $p = 0.08$; or RSImod (0.53 ± 0.14 m·s-1), $r(72) = -$

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0.151, p = 0.199. The magnitude of LMM asymmetry did not influence CMJ performance or asymmetry magnitude. Reducing CMJ kinetic asymmetry may require targeted programming, especially for the eccentric loading phase.

Practical Applications - Optimizing body composition for player health and performance is commonplace for strength and conditioning specialists. This includes developing, maintaining, or restoring body composition symmetry. In addition, results of this study indicate that interlimb differences in LMM are not associated with the differences in peak force and impulse output during a CMJ. Thus, while body composition management should continue, it cannot be assumed that LMM symmetry translates to equal performance by both limbs during a bilateral task. As results of this study show, even highly skilled athletes familiar with a CMJ possess asymmetries that may affect performance. Higher jumping is predicated on the equal application of force in the eccentric and concentric phases. Those athletes with an equal application of force jumped higher than those who did not have an equal application of force in the jump during performance. This may be related to neuromuscular activation and coordination instead of LMM symmetry alone. This was especially evident for the eccentric loading phase of the CMJ where the athlete decelerates their COM. Practitioners may need to train symmetrical force and power production, especially in the eccentric phase of the CMJ. For players with low CMJ performance (height) and greater LMM asymmetry, the practitioner may implement unilateral strength exercises targeting the weaker leg to attempt and reduce the asymmetry. For players with low CMJ performance and little to no asymmetry, practitioners may select more bilateral power exercises to increase jump performance. Practitioners may also consider short-term neuromuscular training (NMT) in training programs to allow these athletes to improve jump performance metrics with a focus on the eccentric phase. Short-term NMT programs will allow the athlete to potentially improve jump landing mechanics and enhance performance in explosive jumping tasks. Implementation of jump training tasks paired with sprint training tasks and strength training tasks over 6 weeks in 1-hour sessions can lead to improvements for jump performance metrics. Other NMT techniques should be focused on the eccentric deceleration or loading phase as there is greater neuromuscular control throughout the eccentric phase compared with the concentric phase. The use of force plates allows for the monitoring of asymmetry during all phases of the CMJ and would be useful to identify deficits, individualize training, and monitor progress.

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Multiscale hamstring muscle adaptations following 9 weeks of eccentric training. Journal of Sport and Health Science, Ahead of Print: 1-10, 2024.

Background: Eccentric training, such as Nordic hamstring exercise (NHE) training, is commonly used as a preventive measure for hamstring strains. Eccentric training is believed to induce lengthening of muscle fascicles and to be associated with the addition of sarcomeres in series within muscle fibers. However, the difficulty in measuring sarcomere adaptation in human muscles has severely limited information about the precise mechanisms of adaptation. This study addressed this limitation by measuring the multiscale hamstring muscle adaptations in response to 9 weeks of NHE training and 3 weeks of detraining.

Methods: Twelve participants completed 9 weeks of supervised NHE training, followed by a 3-week detraining period. We assessed biceps femoris long-head (BFlh) muscle fascicle length, sarcomere length, and serial sarcomere number in the central and distal regions of the muscle. Additionally, we measured muscle volume and eccentric strength at baseline, post-training, and post-detraining.

Results: NHE training over 9 weeks induced significant architectural and strength adaptations in the BFlh muscle. Fascicle length increased by 19% in the central muscle region and 33% in the distal muscle region. NHE also induced increases in serial sarcomere number (25% in the central region and 49% in the distal region). BFlh muscle volume increased by 8%, and knee flexion strength increased by 40% with training. Following 3 weeks of detraining, fascicle length decreased by 12% in the central region and 16% in the distal region along with reductions in serial sarcomere number.

Conclusion: Nine weeks of NHE training produced substantial, region-specific increases in BFlh muscle fascicle length, muscle volume, and force generation. The direct measurement of sarcomere lengths revealed that the increased fascicle length was accompanied by the addition of sarcomeres in series within the muscle fascicles.

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Platelet-poor plasma for the treatment of acute hamstring muscle injuries in collegiate football athletes: A cohort study. Clinical Journal of Sport Medicine, Ahead of Print: 1-5, 2024.

Objective: To evaluate the efficacy of platelet-poor plasma injections for the treatment of acute hamstring muscle strains.

Design: Cohort study.

Setting: Outpatient Sports Medicine Clinic.

Participants: Twenty consecutive patients with an acute hamstring muscle injury from 2021 to 2022 were included.

Interventions: Platelet-poor plasma injection into the hamstring muscle strain.

Main outcome measures: Time (days) until return to full unrestricted participation in sport.

Results: All patients were male collegiate football athletes, with an average age of 20 (±1.6) years old. The most commonly injured muscle was the semitendinosus (65%). After treatment with platelet-poor plasma, patients on average returned to full, unrestricted participation in sport at 29.4 (±7.2) days postinjury, with a statistically significant improvement in pain as early as 2 weeks postinjection. No recurrent injuries to the affected hamstring muscles were seen within 12 months.

Conclusions: PPP may be a safe and effective treatment for acute hamstring muscle strains and may potentially expedite time to return to play.

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Refining the recipe or spoiling the broth? Reframing perceptions of program adaptation in sports injury prevention. British Journal of Sports Medicine, Ahead of Print: 1-2, 2024.

This article discusses the challenges and strategies for implementing injury prevention exercise programs in real-world conditions. Here's a summary:

Key Points:

1. **Efficacy vs. Implementation**: While many injury prevention programs are effective in controlled trials, implementing them in real-world settings is challenging. Teams often modify these programs to fit their specific needs, which can affect the program's fidelity (how closely it follows the original design).
2. **Adaptation-Fidelity Dilemma**: There's a balance between adapting programs to fit specific contexts and maintaining fidelity to the original protocol. Strict adherence to the original protocol can ensure desired outcomes but may not consider the unique needs of each team.
3. **RE-AIM Framework**: Researchers use the RE-AIM framework to evaluate implementation outcomes, focusing on Reach, Effectiveness, Adoption, Implementation, and Maintenance. However, most analyses don't account for program modifications, which are common in practice.
4. **Copenhagen Adduction Exercise (CAE)**: Under controlled conditions, the CAE program significantly reduced groin problems. However, real-world adherence to the original protocol was low. Many teams modified the program while retaining its core components, leading to higher adherence rates.
5. **Hybrid Approach**: The article suggests moving beyond a one-size-fits-all approach and embracing a hybrid method. This involves maintaining key elements of the program while allowing for context-specific modifications. This approach can enhance adherence and effectiveness.

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Conclusion:

The article advocates for a dynamic, iterative process in developing and implementing injury prevention programs. Researchers and practitioners should work together to identify core components and allow for thoughtful modifications to fit specific contexts. This can ultimately benefit athletes by reducing injury risks.

Repeated sprint training: The effects of session volume on acute physiological, neuromuscular, perceptual and performance outcomes in athletes. European Journal of Sport Science, Ahead of Print: 1-13, 2024.

Abstract - We examined the effects of repeated sprint training (RST) session volume on acute physiological, neuromuscular, perceptual and performance outcomes. In a randomised, counterbalanced, and crossover design, 14 healthy and trained male and female athletes (age: 23 ± 3 years) completed two sets of 10 × 40 m (10 × 40), 5 × 40 m (5 × 40), 10 × 20 m (10 × 20) and 5 × 20 m (5 × 20) sprints with 30 s rest between repetitions and 3 min rest between sets for all protocols. Average and peak heart rate, average oxygen consumption (VO2), time >90% of maximal oxygen consumption (VO2max), differential ratings of perceived exertion (RPE), session-RPE training load (sRPE-TL), percentage sprint decrement (Sdec), acceleration load and distance >90% of maximal sprint speed were recorded during each session. Neuromuscular performance (i.e. countermovement jump, lower-limb stiffness and isometric hamstring strength) were measured post-session, 24 h, and 48 h and compared to pre-session. A univariate analysis of variance was used to compare within- and between-protocol differences. To aid data interpretation, all effects were expressed as an effect size and accompanied by probability values (pMET). The 10 × 40 protocol induced the greatest training load compared to all other protocols (pMET < 0.05), including moderate to very large differences in breathlessness RPE, large differences in Sdec and time >90% VO2max and very large differences in sRPE-TL. The 5 × 20 protocol induced the lowest training load compared to all other protocols (pMET < 0.05), including moderate to large differences in sRPE-TL and leg muscle RPE. Heart rate, VO2, sRPE-TL, leg muscle RPE and Sdec were similar between 5 × 40 and 10 × 20 (pMET < 0.05), but the acceleration load was greater for 10 × 20 when compared to 5 × 40 (pMET < 0.001), and this difference was large. Changes in neuromuscular

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performance across all timepoints and all protocols were unclear. Larger session volumes increase the demands of RST and by manipulating volume, sprint distance and the number of repetitions, practitioners can alter the internal and external training load.

Highlights

- The 10 X 40 protocol (i.e. 800 m volume) induced the greatest physiological and perceptual demands, which was demonstrated by a session average heart rate and VO2 of 89% ± 3% and 72% ± 8% of max, respectively, whereas differential RPE were rated hard to very hard on average.
- Performing 10 repetitions did not substantially increase the average VO2 demands when sprint distance was matched (e.g. 10 X 40 vs. 5 X 40). Rather than prescribing high repetition protocols, practitioners are encouraged to increase sprint distance if a greater aerobic stimulus is desired.
- There were no differences in the heart rate and VO2 recovery between the 2nd and 3rd minute of interset recovery. Therefore, to reduce session duration while providing adequate recovery of cardiorespiratory function, 2min passive rest periods can be prescribed.

Return to play after shoulder stabilization – What if this isn't going well? Operative Techniques in Sports Medicine, Ahead of Print: 1-24, 2024.

Abstract- Traumatic shoulder instability is prevalent among athletes, often leading to surgical intervention and challenging rehabilitation outcomes. This chapter explores the complexities surrounding return to play (RTP) following shoulder stabilization procedures, highlighting the significant variance in RTP rates and the common lag in returning to preinjury performance levels, particularly in throwing athletes. Factors influencing successful outcomes include the nature of the instability, surgical approach, and individual athlete characteristics. A consensus on RTP criteria remains elusive, with a predominant reliance on time-based recovery, yet evidence suggests that other physical and psychological variables must be considered. Emphasizing a shared decision-making process and ongoing assessment of range of motion, strength, and proprioception, this chapter aims to identify key deficits and interventions that can enhance rehabilitation success for athletes struggling to return to play.

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Effective communication among the sports medicine team and the athlete is essential to optimize adherence to rehabilitation protocols and improve overall outcomes.

Return to play in professional football players following traumatic cervical spine injury: Expert opinions from the National Football League spine surgeons. JNS Spine, Online: Dec. 20: 1-10, 2024.

Objective: There is a paucity of high-quality return-to-play (RTP) data following treatment of cervical spine injuries in contact sports. In this study, the authors gathered insights from National Football League (NFL) team spine surgeon consultants to highlight current practices in treating cervical spine injuries and report decision-making regarding RTP in professional American football players.

Methods: A cross-sectional, online survey was distributed to all NFL consulting physicians specializing in the management of spine injuries. The survey covered the following five clinical vignettes of cervical spine injuries: 1) radiculopathy, 2) myelopathy, 3) unilateral facet fracture, 4) unilateral facet dislocation, and 5) neck pain with MRI showing a ligamentous STIR signal. Participants were asked about management options and criteria to clear players for RTP using a combination of multiple-choice and open-ended answers.

Results: A total of 26 physicians from 21 of 32 (66%) teams responded. Anterior surgery was most commonly recommended for cervical disc herniation causing radiculopathy or myelopathy (73% and 88%, respectively). A rigid cervical orthosis was preferred by 68% of experts for initial management of nondisplaced unilateral facet fracture, but single-level anterior fusion was preferred by 56% for a unilateral facet dislocation. Common criteria to clear players for RTP with cervical disc herniation causing radiculopathy included a normal examination (85%), radiographic fusion postoperatively (58%), and pain-free range of motion (50%). Contraindications for RTP included persistent stenosis (35%), instability (31%), multilevel fusion (27%), and persistent cord signal change (23%). Additional criteria for RTP in players with facet fractures or dislocations included radiographic evidence of fracture healing (32% and 24%, respectively) and normal flexion/extension radiographs (24% and 32%, respectively). Finally, for players with isolated ligamentous STIR signal changes, resolution of MRI findings was required by 36% of responding physicians prior to RTP.

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Conclusions: Decision-making regarding RTP after cervical spine injuries in professional football players is complex and influenced by improvement of symptoms, pain-free range of motion, and radiographic evidence of fusion or fracture healing. Respondents preferred anterior cervical discectomy and fusion for disc herniations causing cervical radiculopathy and myelopathy, rigid orthosis for unilateral facet fractures, and surgery for unilateral facet dislocations. The results of this study provide insight into how surgeons serving as consultants to professional football teams may counsel players who sustain cervical spine injuries.

Risk and prognosis of hamstring injuries in the National Football League: A 12-year review. The Orthopaedic Journal of Sports Medicine, 12(12): 1-5, 2024.

Background: Hamstring injuries can have a significant burden on a professional football player's career and performance. One in 4 players will suffer a lower extremity strain in their career, with the hamstring being the most common. These injuries are often fraught with extended periods of recovery and reinjury rates varying between 16% and 38%.

Purpose: To determine the progression and duration of hamstring injury as well as risk factors and probability of reinjury in the National Football League (NFL).

Study design: Descriptive epidemiology study.

Methods: Data were extracted from the official publicly accessible NFL database of injuries. NFL players suffering a hamstring injury during the study period of 2008-2020 were identified. Injuries were reported weekly as mandated by the NFL. Players were deemed cleared from injury when no longer listed in the injury report (IR). Reinjury was defined as any subsequent hamstring injury after a player's initial injury. The duration of injury and factors associated with injury were analyzed. Secondary objectives included team-specific injury rates and timing of injury.

Results: A total of 2101 hamstring injuries occurred among 1354 players. The mean duration of the IR was 2.4 ± 2.1 weeks. A total of 442 (33%) players suffered subsequent reinjury, 27% of which occurred during the same season. The mean duration on the IR for reinjury was 2.4 ± 1.8 weeks, with no significant difference in duration compared with initial injury (2.4 ± 1.8 vs 2.2 ± 1.7; P = .138). Defensive

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skill players were most frequently injured (50.4%) followed by offensive skill players (37.7%). Of the 4812 total IR listings, 1806 (38%) did not participate in practice. A total of 1590 (33.5%) had limited participation in practice, and 1353 (28.5%) had full participation in practice. Injury rates among teams varied and ranged from 0.6% (Baltimore Ravens) to 5.7% (Houston Texans), with a median of 3.1% ($P < .0001$).

Conclusion: Hamstring injuries can have a lasting impact on a player's career with more than a third of players suffering subsequent reinjury. Although suffering a hamstring injury predisposes a player to reinjury, previous injury was not associated with longer recovery times. Rates of injury among organizational teams vary; thus, further research to understand the difference in preventive measures and treatment protocols may be warranted.

The effect of multi-ingredient protein versus collagen supplementation on satellite cell properties in males and females. Medicine and Science in Sports and Exercise,56(11): 2125-2134, 2024.

Introduction: Skeletal muscle satellite cells (SC) contribute to the adaptive process of resistance exercise training (RET) and may be influenced by nutritional supplementation. However, little research exists on the impact of multi-ingredient supplementation on the SC response to RET.

Purpose: We tested the effect of a multi-ingredient supplement (MIS) including whey protein, creatine, leucine, calcium citrate, and vitamin D on SC content and activity as well as myonuclear accretion, SC and myonuclear domain compared with a collagen control (COL) throughout a 10-wk RET program.

Methods: Twenty-six participants underwent a 10-wk linear RET program while consuming either the MIS or COL supplement twice daily. Muscle biopsies were taken from the vastus lateralis at baseline and 48 h after a bout of damaging exercise, before and after RET. Muscle tissue was analyzed for SC and myonuclear content, domain, acute SC activation, and fiber cross-sectional area (fCSA).

Results: MIS resulted in a greater increase in type II fCSA following 10 wk of RET (effect size (ES) = 0.89) but not myonuclear accretion or SC content. Change in myonuclei per fiber was positively correlated with type I and II and total fiber hypertrophy in the COL group only, indicating a robust independent

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effect of MIS on fCSA. Myonuclear domain increased similarly in both groups, whereas SC domain remained unchanged following RET. SC activation was similar between groups for all fiber types in the untrained state but showed a trend toward greater increases with MIS after RET (ES = 0.70).

Conclusions: SC responses to acute damaging exercise and long-term RET are predominantly similar in MIS and COL groups. However, MIS can induce greater increases in type II fCSA with RET and potentially SC activation following damage in the trained state.

The effect of the Copenhagen Adduction and Nordic Hamstring exercises on muscle strength and athletic performance: A quasi-experimental study. Journal of Orthopaedic Reports, Ahead of Print: 1-42, 2024.

Background: Two of the most common exercises used in prevention as well as rehabilitation protocols are the Nordic Hamstring (NH) and Copenhagen Adduction (CA). Although their strength effect has been investigated, the effect of the combination of both exercises on muscle strength, jump and sprint performance of male football (soccer) players has yet to be determined.

Materials and Methods: A total of 38 players (15.3±0.49 years, 59.7±7.14 kg, 152.1±53.3 cm) of a Greek football club participated in a progressive 4-week strengthening program, performing both the NH and CA exercises twice a week during the team's pre-season period. Isometric hamstring and eccentric hip adduction strength were assessed using a hand-held dynamometer, while jump and sprint performance were also tested. Perceived exertion was recorded throughout the intervention.

Results: Isometric hamstring strength increased by 11% ($p<0.000$) and eccentric adduction strength increased by 17% ($p<0.000$). There was a decrease of 5-7% ($p=0.000$ to $p=0.034$) in jump performance, and an improvement of 2-3% ($p=0.009$ to $p=0.253$) in sprint performance. There was no statistically significant difference for perceived exertion between the two exercises ($p=0.084$-1.000).

Conclusion: A 4-week progressive exercise program with the Nordic Hamstring and Copenhagen Adduction exercises combined resulted in statistically significant improvements in both isometric hamstring and eccentric adduction strength, but it did not lead to significant results in jump and sprint performance.

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The effect of the Nordic Hamstring exercise on hamstring muscle activity distribution during high-speed running estimated using multichannel electromyography: A pragmatic randomized controlled trial. Clinical Journal of Sport Medicine, Ahead of Print: 1-10, 2024.

Objective: To evaluate the effect of the Nordic hamstring exercise on normalized muscle activity and relative contribution of the biceps femoris long head, semitendinosus, and semimembranosus through multichannel electromyography in the late-swing phase of high-speed running.

Design: A pragmatic, 2-arm, single-center randomized controlled trial. Participants were randomly assigned to a Nordic group or control group.

Setting: Dutch male basketball.

Participants: Twenty injury-free players (mean age 18 ± 3 years).

Intervention: A 12-week Nordic hamstring exercise intervention.

Main outcome measures: Level of normalized muscle activity (percentage maximal voluntary isometric contraction [%MVIC]) and relative contribution (%con) of hamstring muscles for 12 weeks.

Results: The Nordic hamstring exercise intervention did not result in significant changes for 12 weeks. For normalized muscle activity, between-group differences (compared with the control group) for 12 weeks were 11.4 %MVIC (95% confidence interval [95% CI]: -11.0, 33.8) for the biceps femoris long head, -9.4 %MVIC (95% CI: -23.3, 5.2) for the semitendinosus, and -2.7 %MVIC (95% CI: -15.8, 10.3) for the semimembranosus, P = 0.151. For relative contribution, between-group differences for 12 weeks were -6.1 %con (95% CI: -2.4, 14.6) for the biceps femoris long head, -7.0 %con (95% CI: -13.6, -0.4) for the semitendinosus, and 0.9 %con (95% CI: -9.2, 11.0) for the semimembranosus P = 0.187. Positive values are in favor of the Nordic group.

Conclusions: A 12-week Nordic hamstring exercise intervention did not affect the level of muscle activity and relative contribution of hamstring muscles in the late-swing phase of high-speed running. Because of the low amount of data sets, results should be interpreted cautiously.

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The effectiveness of cryostimulation exposure on sleep and recovery in male athletes: Timing considerations. European Journal of Sport Science, Ahead of Print: 1-10, 2024.

Abstract - The objectives of the study were (i) to evaluate the effects of whole-body cryostimulation (WBC) after training as an effective tool to improve sleep and recovery and (ii) to determine which exposure schedule for WBC (immediately after training vs. 1h before bedtime) is the most appropriate to improve sleep and recovery. Twenty-two well-trained male athletes or triathletes (23.1 ± 3.5-years) were recruited in this controlled crossover protocol. Each participant completed the three experimental sessions, each composed of a running training session and one night of sleep analysis. Depending on the experimental condition, subjects were exposed-or not (CONT)-to WBC immediately after training (CRYO 7 p.m.) or 1h before going to bed (CRYO 10 p.m.). All participants were randomly assigned to each condition, spaced seven days apart. Sleep was monitored via actimetry, cerebral activity recordings, heart rate variability, body core temperature, and perceptual measurements. WBC after training did not modify subjective or objective sleep parameters compared to the control condition. However, WBC significantly improved cardiac parasympathetic activity. More specifically, it was more activated during the night following the CRYO 10 p.m. condition than in the other experimental conditions. Core body temperature (CBT) 1h after bedtime was the lowest in the CRYO 10 p.m. condition. This study confirmed the effectiveness of WBC as a recovery technique after intense evening training. We suggest that WBC may exhibit the most significant benefit on recovery if the session is performed 1h before bedtime, acting on subsequent parasympathetic activation and CBT drop.

Highlights

- Cryotherapy improves sleep through better parasympathetic activation.
- Time of exposure affects sleep parameters differently.
- An exposure to cryostimulation close to sleep (at 10 p.m.) improves parasympathetic activation even more than an exposure just after training (at 7 p.m.).
- An exposure to cryostimulation close to sleep (at 10 p.m.) decreases the internal temperature to a greater extent than an exposure just after training (at 7 p.m.).

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The potential role of retinal slip on National Football League (NFL) officiating and its effect on on-field decision making: Mechanisms and countermeasures. The Royal College of Ophthalmologists, Ahead of Print: 1-6, 2024.

Abstract - Visual processing involves numerous neuro-ophthalmic pathways that enable humans to perceive and interact with the world. These visual pathways can negatively impact visual perception and decision-making. The Vestibulo-Ocular Reflex (VOR) is a compensatory mechanism that stabilizes gaze on a target during head movement, such as running or turning. However, inadequate VOR stabilization can lead to retinal slip, reducing visual acuity. In the National Football League (NFL), officials require exceptional visual training and performance. NFL officiating utilize rapid visual processing, strategic movements, and precise judgement to make accurate penalty calls in real-time. Retinal slip can adversely affect play calling and penalty assessment by NFL officials. Despite these challenges, NFL officials achieve high precision and accuracy in up to 98.9% of plays. To maintain such high standards, the NFL has implemented a standardized curriculum to mitigate the effects of VOR and retinal slip. This manuscript outlines the basics of VOR function and explores countermeasures to minimize retinal slip in NFL officiating. We believe that these scientific endeavours can improve both penalty assessment accuracy and precision. The use of these general principles might also be potentially applicable to other sports.

The role of speed, change of direction, and momentum by position and starting status in Division 1 collegiate football players. Journal of Strength and Conditioning Research, 39(1): 41-47, 2024.

Abstract - This study (a) investigated differences between big, mid, and skill positions in sprint and change of direction times and momentum; (b) compared starting and nonstarting athletes; and (c) investigated whether thresholds can be developed to distinguish between starting and nonstarting Division 1 collegiate football athletes. Data from 496 collegiate football players who completed the 40-yard dash, pro-agility, and L drill were analyzed. Momentum was calculated using body mass and the average velocity during each test. To assess differences between positions and starters and nonstarters, data were analyzed using linear mixed models with effect size ±95% confidence intervals. Receiver

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operating characteristic (ROC) curves were generated to determine whether a cutoff value could be used to distinguish starters from nonstarters. Significant differences for both time and momentum were found between positional groups and starters and nonstarters for all tests in all positions. Starting skill position players tended to have greater differences in sprint or change of direction times and starting big players had greater sprint momentum. However, it should be noted that all ROC curves demonstrated relatively poor predictive value. Collectively, these findings demonstrate that bigger, faster players are preferentially selected in collegiate Division 1 football and there may be value in coaches collecting and assessing different outcome measures (e.g., sprint times and sprint momentum) depending on the positional group of the player. Finally, it should be acknowledged that setting binary thresholds to guide selection decisions is ill-advised and that speed, change of direction, and momentum are only one piece of the performance puzzle.

Practical Applications - American football athletes are required to be fast, change direction efficiently, and execute these skills with substantial momentum. However, different positional groups require differing levels of these qualities. For example, skill players should focus on absolute speed and change of direction performance. Alternatively, big- and mid-position athletes need to have significant body mass to complement their running speed so that they have adequate momentum during collisions. Thus, it is recommended that the calculation of momentum during sprint and change of direction tasks is calculated to support coaches. In addition, it is acknowledged that starting players tend to have greater speed, change of direction, and momentum qualities when compared with nonstarters. Therefore, developing these qualities may be desirable and the data provided within this manuscript can be used as a reference for coaches and athletes. However, using thresholds to support selection decisions or identify potential talent is ill-advised because of the multifaceted nature of American football. Instead, the development of speed, change of direction, and momentum should be prioritized for all players, with skill positions emphasizing absolute speed and change of direction, big positions ensuring they have adequate body mass to allow for momentum to be created, and midposition players carefully balancing the development of both speed and body mass.

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The state of the science for potential contributors to musculoskeletal injury following concussion: Mechanisms, gaps, and clinical considerations. Musculoskeletal Science and Practice, 75: 103219, 2025.

Abstract - Concussion is a common mild traumatic brain injury affecting athletic, military, and general populations. While unrestricted medical clearance often occurs within 1-month, emerging evidence indicates prolonged sensorimotor control deficiencies and greater musculoskeletal injury risk after a concussion. Various theoretical frameworks and original studies indicate potential neurophysiological alterations affecting sensory input, sensorimotor integration, and neuromotor output, though the exact mechanisms remain unclear. This commentary aims to briefly review the literature on contributors to sensory and neuromotor deficiencies potentially related to musculoskeletal injury risk, recommend future research avenues to better understand these factors, and highlight current and future clinical applications.

The structure, function, and adaptation of lower-limb aponeuroses: Implications for myo-aponeurotic injury. Sports Medicine – Open, 10: 133, 2024.

Abstract - The aponeurosis is a large fibrous connective tissue structure within and surrounding skeletal muscle and is a critical component of the muscle-tendon unit (MTU). Due to the lack of consensus on terminology and the heterogeneous nature of the aponeurosis between MTUs, there are several questions that remain unanswered. For example, the aponeurosis is often conflated with the free tendon rather than being considered an independent structure. This has subsequent implications when interpreting data regarding the structure, function, and adaptation of the aponeuroses from these studies. In recent years, a body of work has emerged to suggest that acute injury to the myo-aponeurotic complex may have an impact on return-to-sport timeframes and reinjury rates. Therefore, the purpose of this review is to provide a more detailed understanding of the morphology and mechanical behaviour common to all aponeuroses, as well as the unique characteristics of specific lower-limb aponeuroses that are commonly injured. This review provides the practitioner with a current understanding of the mechanical, material, and adaptive properties of lower limb aponeuroses and suggests directions for future research related to the myo-aponeurotic complex.

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Theorizing the use of performance enhancing substances and methods in sport as four different types of deviant behavior. Deviant Behavior, Ahead of Print: 1-19, 2024.

Abstract - The central objective of this paper is to demonstrate that the validity of the Hughes and Coakley (H&C) model of deviance in sport is context specific and depends on the time, place, social groups involved, and the relative power of the audience and deviant(s). H&C argued that performance enhancing substance (PES) use constituted a single type of deviant behavior among athletes (i.e. positive deviance). The Heckert and Heckert (H&H) deviance framework made it possible to theorize performance-enhancing substances and methods (PESM) use as four ideal types of deviant behavior among athletes and in sports collectives. Given the variability in the historical and social contexts where PESM use has occurred in sport, a comprehensive explanatory framework is needed to understand a full range of PESM behaviors. This paper demonstrates that the H&H framework has a clear explanatory advantage over the H&C model.

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